JOSHUA NDAKOTSU

Microbiologist | Environmental Microbiologist | Soil Scientist
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Computational biology/bioinformatics portfolio: jndakotsu.github.io

PROFESSIONAL SUMMARY

Soil scientist specializing in soil microbiology and environmental systems, with formal training in soil mapping, land evaluation, soil fertility assessment, and organic matter management. Experienced in soil microbial ecology, soil health, water quality analysis and soil-plant-microbe interactions, with proficiency in a range of laboratory and field-based techniques including microbial enumeration, enzyme activity assays, PCR, physicochemical characterization, and organic waste stabilization. Research background spans soil quality assessment, bioremediation, and soil amendment evaluation, underpinned by strong analytical and experimental design skills developed through undergraduate, postgraduate and professional training.

CORE COMPETENCIES

Soil microbial ecology, PGPB isolation & biofertilization, Molecular techniques (PCR, DNA Extraction, Electrophoresis), Soil enzyme assays, Microbial enumeration & culturing, Soil physicochemical characterization, Organic amendment evaluation, Bioremediation, Water quality analysis, Phytochemical extraction, Antioxidant analysis, Soil mapping & land evaluation, Experimental design, Statistical analysis (GenStat, R), Scientific writing & publication

PROFESSIONAL EXPERIENCE

Soil Scientist (Soil Microbiologist)
Oct 2024 – Present

Institute of Agricultural Research and Training (IAR&T), Ibadan, Nigeria

- Investigate soil-plant-microbe interactions to understand microbial influence on plant growth, soil health and nutrient cycling
- Conduct soil quality assessments including physicochemical characterization and microbial diversity analysis under varied land management practices.
- Apply culture-based and molecular techniques (PCR, DNA extraction, electrophoresis) for soil microbial identification and characterization.
- Contribute to the production of plant growth-promoting bacteria (PGPB) fortified compost for agricultural use.
- Support bioremediation projects focused on microbial degradation of organic and inorganic pollutants.
- Evaluate biofertilization strategies, assessing the roles of beneficial microbes in soil fertility enhancement.
- Perform soil mapping and land evaluation in support of agricultural and environmental research programmes.
- Interpret experimental data and prepare technical reports to inform sustainable agricultural practices.

National Youth Service Corp
Mar 2021 – Feb 2022

Nigerian Nuclear Regulatory Authority (NNRA), Abuja, Nigeria

- Supported routine radiological assessment, environmental sampling, and baseline measurement programmes.
- Conducted investigative analysis of environmental samples for regulatory compliance.

Quality Control Officer
Dec 2020 – Mar 2021

Sam Pharmaceuticals, Ilorin, Kwara State, Nigeria

- Performed microbiological analysis of raw materials, water samples, and finished pharmaceutical products.
- Conducted stability testing and shelf-life evaluation of products.
- Maintained quality control records for each batch manufactured in accordance with GMP standards.

Student Industrial Trainee (SIWES)
Jan 2019 – Jul 2019

Lower Usuma Dam, FCT Water Board, Bwari, Abuja, Nigeria

- Conducted physical, chemical, and bacteriological analysis of water prior to public supply.
- Carried out presumptive and confirmatory microbiological tests to ensure water safety standards.

ADDITIONAL PROFESSIONAL EXPERIENCE

White board animator May 2022 – Oct 2024

Golden Pro Groups Gudu Abuja, Nigeria

Created whiteboard animations for clients at the interval between National service and employment at IAR&T.

EDUCATION

M.Sc. Soil Resources Management (Soil Microbiology) Mar 2025 – Present

University of Ibadan, Ibadan, Nigeria

Thesis: Swine-based amendments as modulators of soil microbial properties and functional dynamics: Implications on micronutrient concentration in maize tissues.

Relevant coursework: Advanced Soil Microbiology, Soil Testing Theory & Practice, Soil Mapping & Land Evaluation, Soil Fertility Assessment & Management, Soil Biochemistry, Soil Organic Matter Management, Soil-Plant-Water Relationships.

B.Sc. Microbiology - Second Class Upper Honours Jul 2016 – Nov 2020

Landmark University, Omu-Aran, Kwara State, Nigeria

Thesis: Bacterial growth inhibition and antioxidant potentials of leaf infusions of Moringa oleifera, Parkia biglobosa, and Vernonia amygdalina

PEER-REVIEWED PUBLICATIONS AND PRESENTATIONS

Akpor, O.B., **Ndakotsu, J**., Evbuomwan, I.O., Olaolu, T.D., & Osemwegie, O.O. (2021). Bacterial growth inhibition and antioxidant potentials of leaf infusions of *Moringa oleifera, Parkia biglobosa*, and *Vernonia amygdalina. Scientific African,* 14 e01001

Ezaka, E., **Ndakotsu, J.,** & Amosun, J. O. (2026). Microbial biofertilizers as a sustainable alternative to inorganic fertilizers and agrochemicals in reversing soil degradation: A review. *Archives of Ecotoxicology, 8*(2), 72–79.

Accepted Manuscripts

Ezaka, E., Adebajo, S.O., Adediran, A.B., **Ndakotsu, J**., & Ukhureigbe, O.M. (2025). Glufosinate ammonium spraying regime and soil bacteria: An assessment of response. *Journal of Plant Science and Crop Production* (Nigeria).

Manuscripts Under Review

Ezaka, E., **Ndakotsu, J.,** & Zubair, A. (2026). Soil Phosphorus Bioavailability and Shifts in Microbial Community Structure in the Era of Climatic Change: A Review. Discover Agriculture

Conference Presentations

Ndakotsu, J., Fagbola, O., Ezaka, E., & Ojo, O. A. (2026). Fortified swine dung amendments as modulators of soil microbial properties and functional dynamics: Implications for micronutrient concentration in maize tissue. *Oral presentation at the 50th Annual Conference of the Soil Science Society of Nigeria (SSSN)*, Ilorin, Kwara State, Nigeria.

RESEARCH EXPERIENCE

Graduate Research Assistant Nov 2024 – March 2026

IAR&T and University of Ibadan, Oyo, Nigeria

- Evaluated soil microbial functional parameters: basal respiration, microbial biomass carbon (chloroform
- fumigation-extraction), urease, and phosphatase activities.
- Conducted comprehensive soil chemical analysis (pH, total N, available P, organic carbon, exchangeable bases) using Kjeldahl, Mehlich-3, Walkley-Black, and AAS methods
- Conducted plant tissue and micronutrient uptake analysis
- Performed microbiological assays including culturable microbial population density and diversity (CFU/g) on Nutrient Agar and Potato Dextrose Agar

- Designed and executed a screenhouse experiment with 10 treatments (air-dried swine dung, swine dung + Tithonia compost, single and dual PGPB-fortified composts, NPK, and control) and 3 replicates
- Analysed temporal changes in bacterial counts and soil respiration in response to herbicide concentration and NPK application
- Assisted in the execution of a screenhouse pot experiment using maize to assess the impact of herbicide spraying frequencies and different doses on soil bacterial communities

Undergraduate Research Assistant Sep 2019 – Dec 2019

Landmark University, Omu-Aran, Nigeria

- Evaluated antibacterial activity of *Vernonia amygdalina, Parkia biglobosa, and Moringa oleifera* leaf infusions using agar diffusion and liquid broth assays.
- Conducted phytochemical screening and antioxidant analysis (DPPH assay, total phenolic content).
- Assisted in the extraction of fungal endophytic secondary metabolites from medicinal plants.

TECHNICAL SKILLS

Soil Microbiology: Soil microbial culturing and enumeration, DNA extraction, PCR amplification, biofortification of composts using Plant Growth Promoting Bacteria, bioremediation of polluted soil

Soil & Plant Analysis: Soil chemical analysis, soil enzyme assays, plant tissue analysis.

Instrumentation & Laboratory Techniques: UV-Vis spectrophotometry and routine laboratory protocols.

Data Analysis & Visualization: Statistical analysis using GenStat and R, data inspection, data management, and visualization in Microsoft Excel.

Experimental Design & Interpretation: Planning, analysing and interpreting experiments.

PROFESSIONAL CERTIFICATIONS

Agripreneurship: Introduction to Agriculture, IT Essentials, Microsoft Technology Associate, Linux Professional Certificate

REFERENCES

Available upon request.